Professional Certificate of Competency in Electrical Power System Fundamentals for Non-Electrical Engineers

Webinar 7

Module 8: Load Forecasting, Planning and Project Evaluation

Module 8 Topics

- Load forecasting and forward planning
- Supply and demand side management
- The smart grid
- Substation automation
- Carbon credit

Need for System Planning

- Optimum utilisation of resources meeting the requirements
- Safety of personnel and trouble free service
- Freedom from unplanned interruptions
- Power quality within acceptable limits
- Optimising the electrical distribution system:
  - Acceptable norms of existing systems
  - Total cost of ownership
    - Correct selection of equipment, a key factor
    - Applicable to new systems and extensions of existing systems

Approach to System Planning

- Collect field level data
- Projection of future growth
- Analyse data for optimised location of key assets
- Decide on basic system parameters
- Decide distribution configuration
- Arrive at equipment ratings/sizing
- Select appropriate equipment
- Ensure maintainability
- Ensure expandability

Factors Impacting Power Demand

The following factors shall be carefully taken into account while forecasting the power need for a new installation and expansions:

- **Load factor:** ratio of average demand of an installation to its peak demand over a period, usually calendar month or day.
- **Diversity factor:** ratio of sum of maximum demand of group of loads to the overall maximum demand of the unit under normal operation.
- **Intermittent loads:** loads that are operated occasionally.
- **Standby loads:** loads (usually in industries) to act as standby to another load to ensure redundancy and continuous operation.

Data Collection - Utility

- Check for new developments
- Make assumptions based on existing consumption patterns of similar services and installations
- Assess new load requirements
- Check for any major demographic patterns in the offing
- Assess seasonal change patterns
- Assess possible growth of loads
  - Use economic indices for growth planning

Data Collection - Industry

- Overall energy requirement and power demand:
  - Indices from similar industries
- List of connected loads and locations
- Pattern of loading (variations due to process requirements):
  - Loads of highly fluctuating nature to be considered separately
- Assess load factor and diversity factor
- Separating critical loads from non-critical
- Assess future growth plans

Typical Studies Needed in Planning

- Load estimation
- Load flow
- Fault level
- Voltage profile
- Motor starting
- Relay coordination
- Stability (transmission and large industrial loads) and transient performance
- Arc flash studies (equipment specific)

Industrial System Study

- Special modules typically adopted using electrical software include:
  - Earthing design
  - Battery sizing
  - Cable sizing
  - Energy management system
    - Automatic in-plant generation control
    - Intelligent feeder and demand management
    - Intelligent load shedding

Load Flow Studies

- Model the network incorporating all known parameters.
- Solve the network for correctness of equipment capacities and unknown parameters.
- Compare the recorded load data to the solved load flow case.
- Study various system configurations.
- Check solved case for:
  - Overloaded cables, overhead lines and transformers
  - Voltage drop and regulation
  - Adjust tap positions on transformers if necessary

Fault Level Studies

- Solve the network for arriving at the fault currents flowing into the system for the design values.
- Compared solved case to information from the utility.
- Study various system configurations.
- Check solved case for:
  - Switchgear being overstressed
  - Underrated cables
  - Switch transformers, lines or cables out to control fault level

Transient Analysis

Transient stability studies:

- Use a solved load flow case as a base
- Apply faults to the system
- Check solved case for:
  - Unstable generators
  - Induction motors that don’t re-accelerate
  - Faulty settings on power swing blocking relays
  - Transient voltage recovery after dips

Transient Analysis

Sub-transient analysis:

- Study the effects of:
  - Lightning on power system
  - Switching capacitor banks
  - Transient torque pulses on motor shafts

Other Studies

- **System reliability:**
  - Study various system configurations, network topologies and abnormal external conditions
- **Protection coordination**

Projection of Future Growth

- A difficult and uncertain exercise
- Needs constant re-evaluation
- Getting right inputs early
- Provide additional capacity (say for a 5 year period):
  - higher investment
- Provide flexibility in the system for additions:
  - lower investment
- Avoid the need for total equipment replacement to accommodate growth by proper initial planning.

Key Assets Location

- Key assets mean major electrical installations:
  - Generating plants
  - Switching/transformer substations
  - Transmission and distribution corridors
- Correct location makes a system optimal:
  - In terms of investment
  - In terms of recurring losses
  - In terms of maintaining proper voltage profile
  - In terms of maintainability and expandability

Configuration

- For industry:
  - Number of incomers
  - Incoming and distribution voltage level
- Redundancy planning:
  - Type of distribution (radial, ring etc.)
  - Type of redundancy
  - Integration of emergency/standby generation
- System earthing:
  - Local statutory requirements
- Equipment for voltage/PF/harmonic control

Selection of Equipment

- Based on equipment available for meeting the required ratings
- Service conditions prevailing at site of installation or use
- Maintenance requirements
- Experience of other users (internal/external)
- Market reputation
- Status of technology
- Standards (national/international)
- Local statutory requirements

Maintainability

- Adequate space for all equipment as planned
- Access for maintenance
- Special means used at site:
  - buildings to be designed for use
- Clearances for safety and maintainability:
  - removal of parts
- Statutory requirements
- Industry standards and guidelines

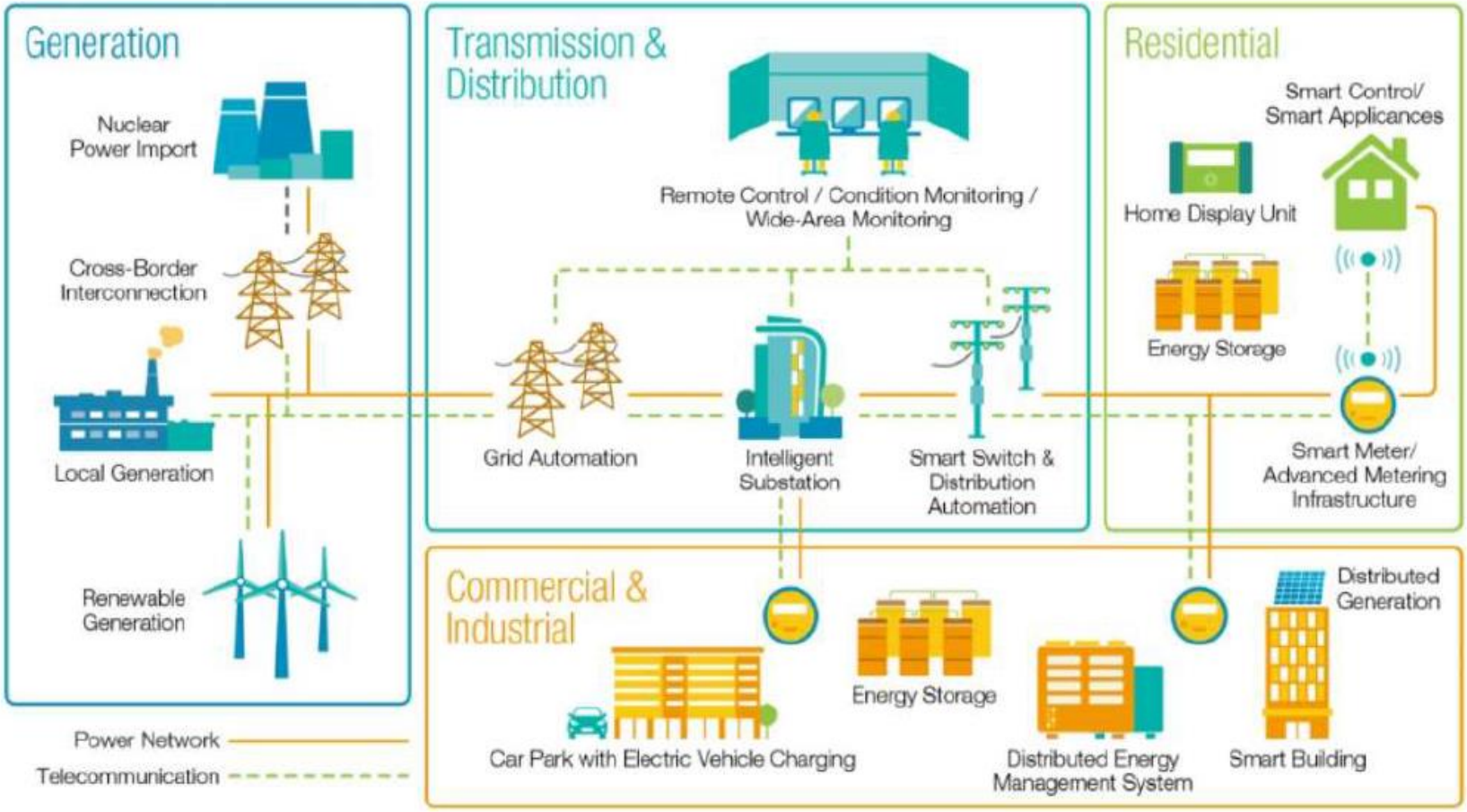
Expandability

- Plan initial equipment in sync with future growth strategies
- Avoid need for major changes
- Additional capacity in-built:
  - Cheaper than expanding later
- Making additions easy:
  - electrically
  - space for additions
- Fault level changes because of expansion

Smart Grid

- Combination of hardware, management and reporting software, with intelligent communications infrastructure.
- Consumers, utility companies have tools to manage, monitor and respond to energy issues.
- Dual conversation between utility and consumer: saving consumers money, energy, delivering more transparency in terms of end-user use, reducing carbon emissions.
- Salient **features:**
  - Transmission optimisation
  - Demand side management
  - Distribution optimisation
  - Asset optimisation

Smart Grid Functions



https://www.elprocus.com/overview-smart-grid-technology-operation-application-existing-power-system/

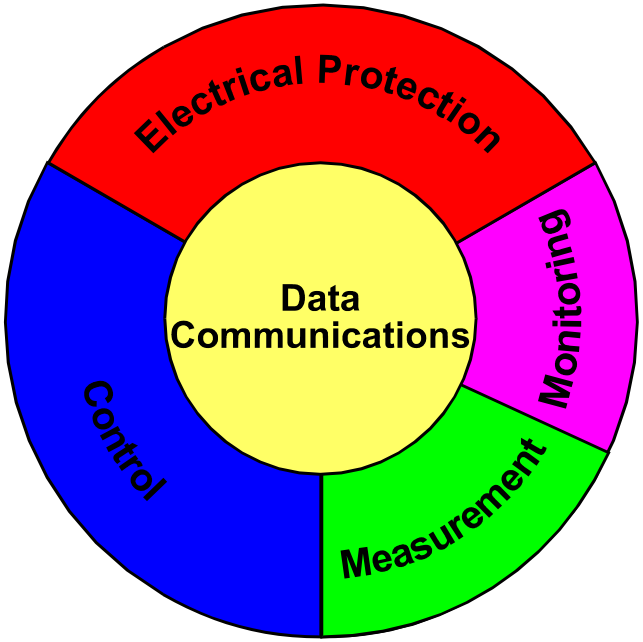
Smart Grid Objectives

- Improved safety and security
- Two way flow of electricity and data
- Maximum efficiency of electricity transmission and distribution networks with reduced losses
- Minimum power disturbances and if any, quick restoration
- Reduction in cost for operation, maintenance, management and ultimately lower power costs for end users
- Minimising the peak demand, which associated lower electricity costs
- Integration of renewable energy systems with large-scale output
- Better metering and asset management

Power System Automation

A system for managing, controlling and protecting a power network.

- Supervisory Control and Data Acquisition (SCADA) has two basic subsystems being part of automation:
  - SCADA master station
  - SCADA remote units
- Has **five key aspects** integrated:
  1. Protection
  2. Control
  3. Measurement
  4. Monitoring
  5. Data communications



Modules and Objectives

- SCADA includes basic modules such as:
  - Data acquisition
  - Control
  - Archiving/database storage
  - The human machine interface (HMI)

Objectives:

- Display real-time data received from remote units
- Keep historical records of data
- Activate alarms when necessary
- Display sequence-of-event reports and disturbance recordings
- Provide an active operator interface
- Perform communication control over the network

Numerical Relay (IED)

- Metering and protection of electrical equipment are integrated into modern numerical relays, also called IEDs (Intelligent Electronic Devices) incorporating:
  - Protection
  - Control and interlocking
  - Measurement
  - Monitoring
  - Communications

Electrical Protection

Electrical protection protects the equipment and personnel, and limits damage in case of an electrical fault.

- Electrical protection is a local function, and should be able to function independently of the power system automation system if necessary, although it is an integral part of power system automation under normal conditions.
- The functions of electrical protection should never be compromised or restricted.
- IEDs as well as other stand-alone protection devices
- Tripping is direct and NOT through automation
- Trip alarms to SCADA
- Trip settings from SCADA

Control and Measurement

Control:

- Local control:
  - Independent of network
  - Configured by user
- Remote control:
  - Operator initiated
  - Network initiated
- Functions:
  - Breaker/isolator switching
  - Interlocking
  - Synchronising checks

Measurement:

- Electrical measurements: voltages, currents, power, power factor, harmonics, etc
- Other analogue measurements:
  - Example transformer and motor temperatures
- Disturbance recordings for fault analysis
- Tariff metering is always by a dedicated device

Monitoring and Communication

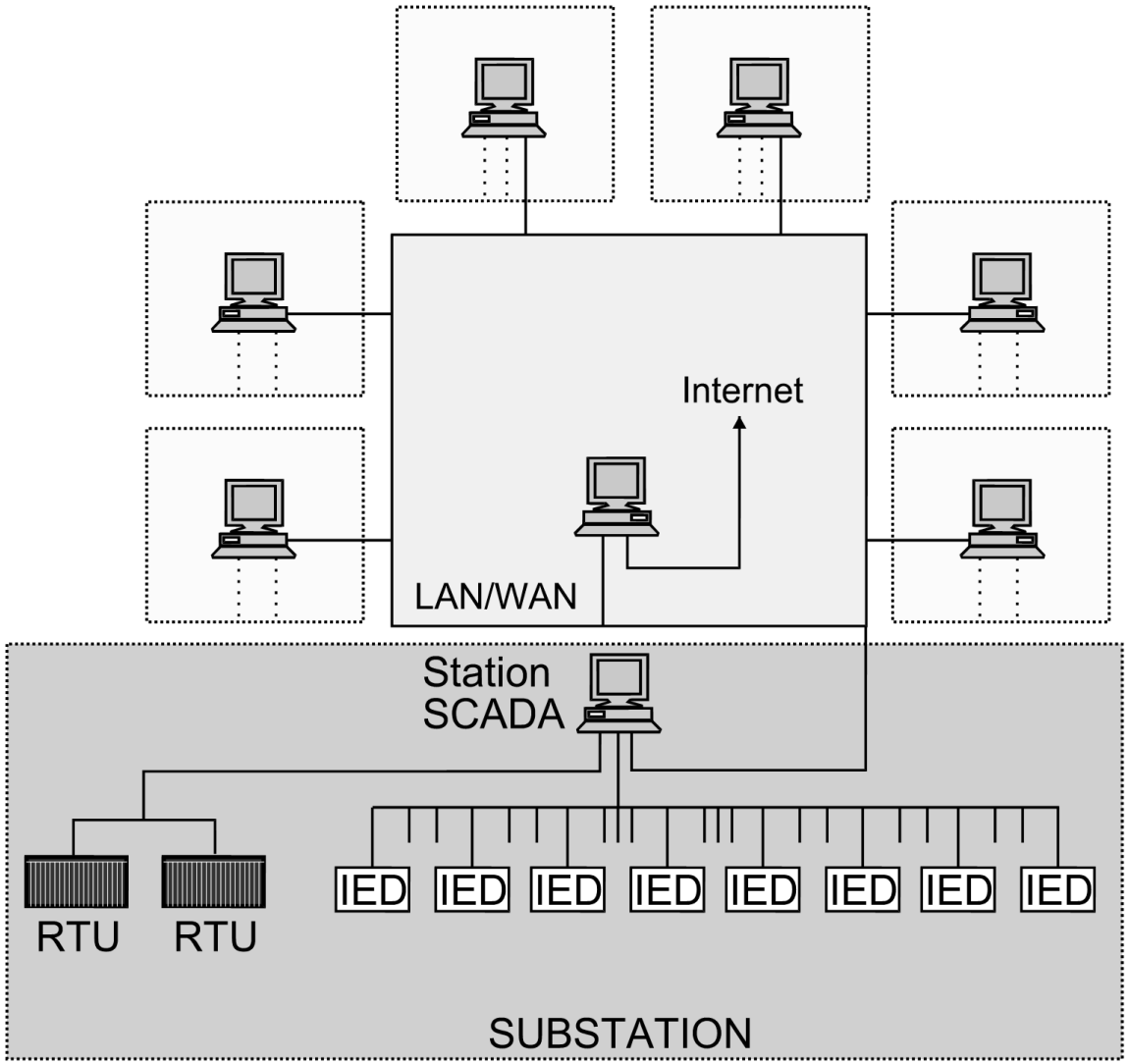
Monitoring:

- Status indications
- Condition monitoring
- Preventative maintenance information
- Sequence-of-event recordings

Data communication:

- Most important
- Loss of communication will not affect:
  - Protection
  - Local control
    - Limited data storage in local devices
  - Remote operation, monitoring, data collection, etc. will not function

SCADA Master at Substation Level



Green House Gas Emissions

- **Burning of fossil fuels:** A major source of industrial emissions
- **Major greenhouse gases (GHGs) emitted:** carbon dioxide, methane, nitrous oxide, hydroflurocarbons.
- Environment gets affected leading to long term issues.
- **Concept of carbon credit:** Came into existence as a result of increasing awareness of need for controlling emissions.
- Mechanism formalised in Kyoto protocol, an international agreement between more than 170 countries.

Environmental Issues

- As more and more energy is produced/consumed by the world population, environmental issues affect the very existence of mankind.
- The energy usage pattern in the last century has affected the world much more than the past centuries.
- **Common issues:**
  - Ozone layer depletion
  - Global warming
  - Loss of biodiversity
- Carbon pollution is the major factor impacting global health.

Kyoto Protocol

Agreement made under the United Nations Framework convention on climate change (UNFCCC).

- Countries ratifying the protocol are committed to reduce their emission of **six major GHGs** given below:

  1. Gases
  2. Carbon Dioxide (CO_2)
  3. Methane (CH_4)
  4. Nitrous Oxide (N_2O)
  5. Sulphur hexafluoride (SF_6)

Carbon Pollution

- It is estimated that 80% of all carbon dioxide emission originates from the combustion of fossil fuels used for energy generation.
- The presence of carbon is also existent in many greenhouse gases exhausted by generators, transport vehicles, refrigerators, air conditioners, etc. viz.:
  - Methane (CH_4)
  - Hydro fluorocarbons (HFCs)
  - Perfluorocarbons (PFCs)

Carbon Credit

- Carbon credits and carbon markets are a component of national and international attempts to mitigate growth of concentrations of greenhouse gases.
- A generic term for any tradable certificate or permit given for a company permitting to emit one ton of carbon dioxide or the mass of another greenhouse gas equivalent to one ton of carbon dioxide
- Companies get a set number of credits as a cap permitted for emission, which declines over time. They can sell any excess to another company.
- Thus, "cap-and-trade" is an incentive to reduce emissions.
- Allows market mechanisms to drive industrial and commercial processes towards low emissions or less carbon intensive avenues.
- Used to finance carbon reduction schemes between trading partners around the world.

End of Webinar 7

Any questions or comments?